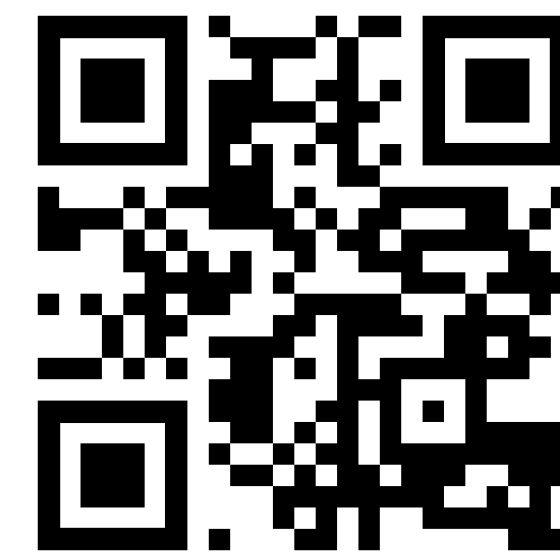


Diagrammatic (∞, n) -categories

Clémence Chanavat

j.w.w Amar Hadzihanovic

Tallinn University of Technology



Atoms and molecules

A **molecule** is combinatorial data encoding the shape of a higher categorical pasting diagram. A molecule U :

- is an **oriented** and graded poset;
- has for each $\alpha \in \{-, +\}$ and $n \in \mathbb{N}$, an **input/output n -boundary**, $\partial_n^\alpha U$, which is a closed subset of U .

Molecules can be **pasted** together: given U , V , and $k \in \mathbb{N}$, if $\partial_k^+ U \cong \partial_k^- V$, then the pushout

$$\begin{array}{ccc} \partial_k^+ U \cong \partial_k^- V & \hookrightarrow & V \\ \downarrow & & \downarrow \\ U & \hookrightarrow & U \#_k V \end{array}$$

is a molecule. Pasting and boundary of molecules satisfy all the axioms of strict ω -categories, and even some **higher-exchange laws** in dimension ≥ 4 . An **atom** is a molecule with a greatest cell.

Directed complexes

A **directed complex** X is a presheaf over the category of atoms. It is the combinatorial data used to construct a **positive** and **regular** polygraphs:

$$\begin{array}{ccc} \coprod_{u \in \mathcal{S}_n} \partial U & \hookrightarrow & \coprod_{u \in \mathcal{S}_n} U \\ (\partial u)_{u \in \mathcal{S}_n} \downarrow & & \downarrow (u)_{u \in \mathcal{S}_n} \\ X_{n-1} & \hookrightarrow & X_n \end{array}$$

There is a construction Mol/X that produces the strict ω -category of **molecules in X** .

- If X is a graph, Mol/X is the free category on X ;
- If X is the n -dimensional simplex, then Mol/X is the n -oriental.

Theorem 1. Under a general acyclicity condition on its molecules, Mol/X is a polygraph, and in fact, a **weak polygraph**.

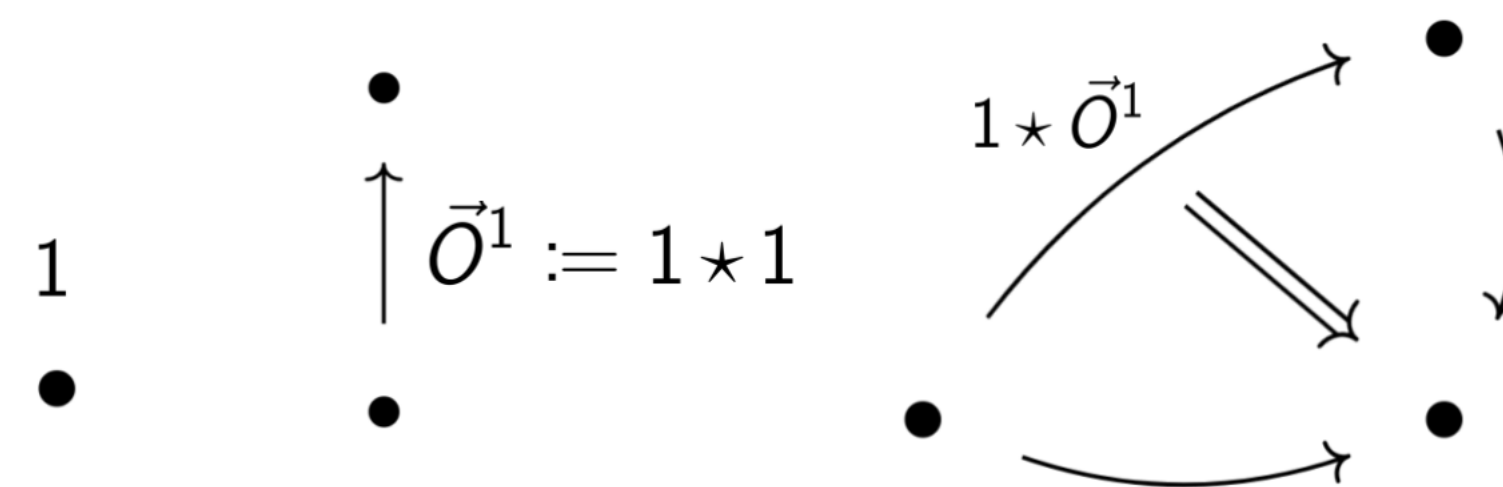
Operations

Atoms, molecules, and directed complexes, natively support higher categorical operations:

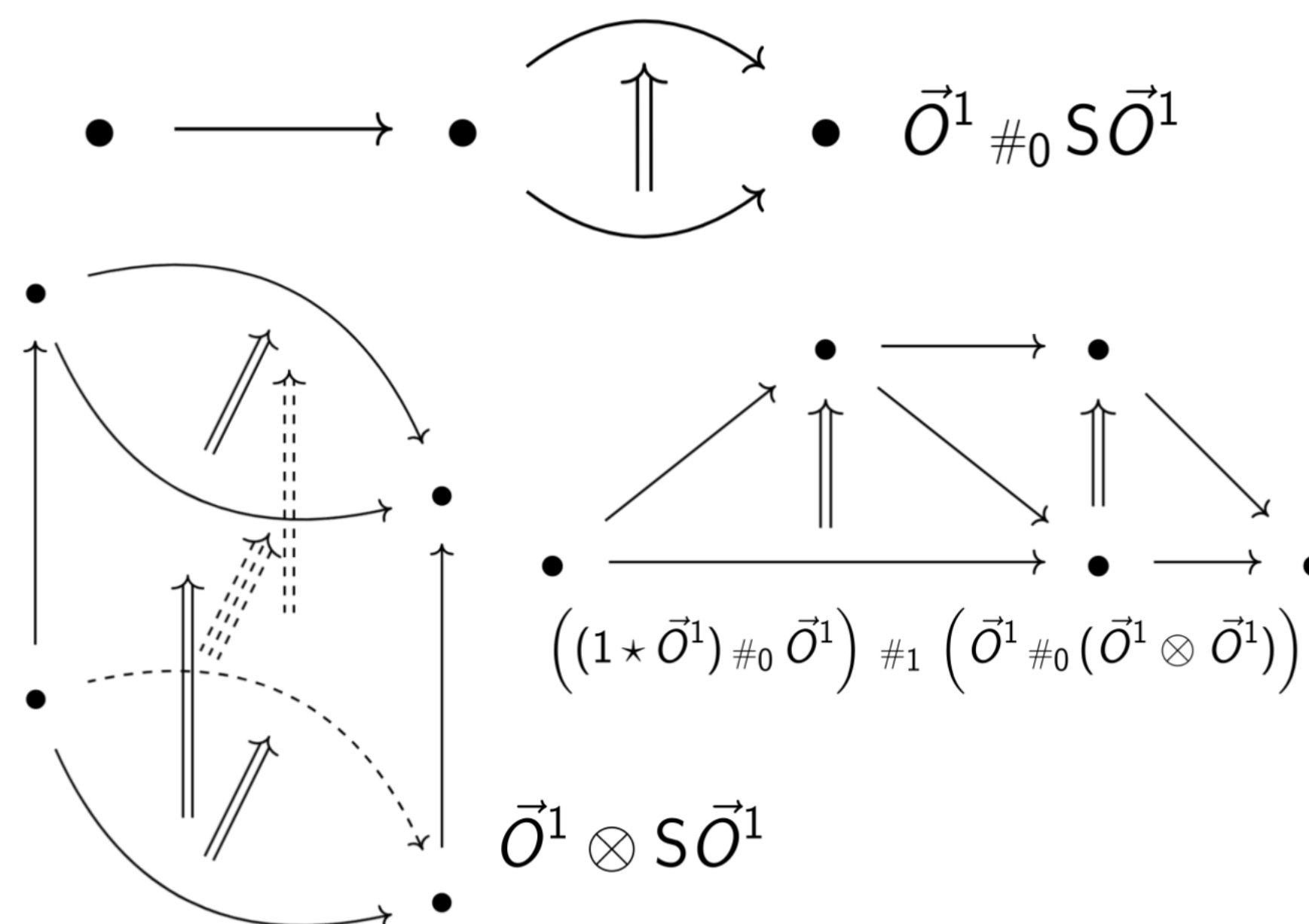
- Gray product $X \otimes Y$,
- join $X \star Y$,
- suspension SX ,
- duals $D_J X$, where $J \subseteq \mathbb{N}$.

Examples

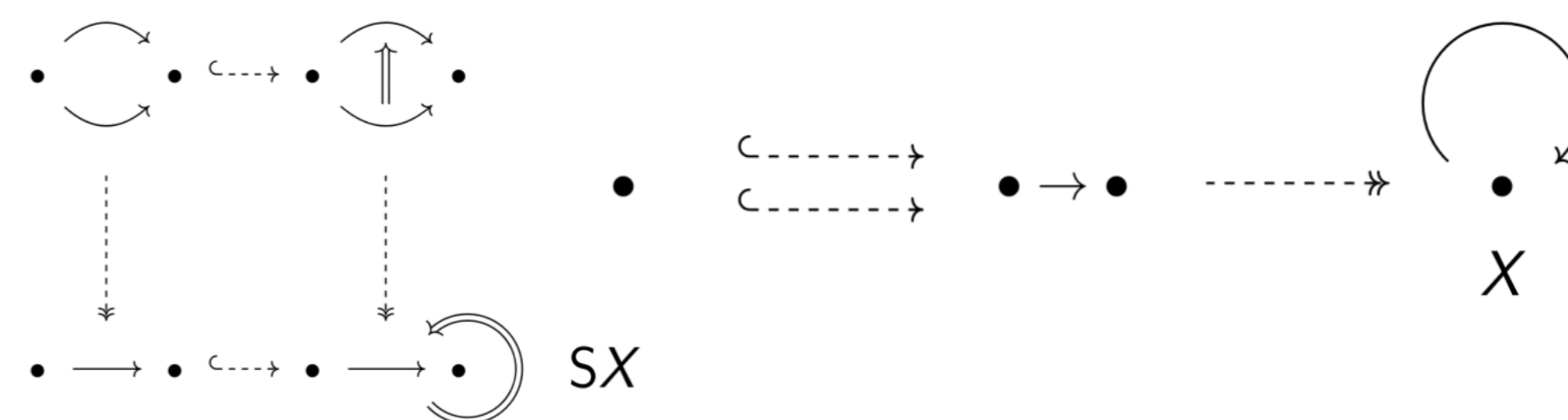
Atoms



Molecules



Directed complexes



In the last example:

- $Mol/X = \mathbf{BN}$
- X and SX satisfy the acyclicity hypothesis of **Theorem 1**.

Diagrammatic (∞, n) -categories

Directed complexes may be used in several ways to construct models of strict n -categories and (∞, n) -categories:

1. as Segal sheaves over the Mol/X 's and **strict functors**;
2. as presheaves over the atoms and **cellular** strict functors between them. A localisation enforces the existence of **weak composites**.

This matches the following two perspectives on standard (∞, n) -categories as

- Segal sheaves over the thetas and strict functors;
- presheaves over the (marked) orientals and cellular strict functors (i.e. over the marked simplex category).

Theorem. Point 1. is a localisation of the (∞, n) -categories at the higher-exchanges, which does nothing when $n \leq 3$.

Conjecture. Point 1. and 2. present the same model of higher categories.

Features of the diagrammatic models

- Very strong **pasting theorems**;
- **Semi-strictification**: marked directed complexes may be used to produce two **equivalent** model:
 1. one where composition is witnessed by **the existence** of weak composites;
 2. one where composition is **an operation** satisfying strict associativity and exchange;

In both cases, the units are algebraic and weak.

- Native support for **higher categorical constructions**, with no need to use different presentations.
- A rich algebra of **weak units**.
- A practical formalism that feels like working with a **topologically sound alternative** to strict n -categories

References

- C. Chanavat and A. Hadzihanovic, *Semi-strictification of (∞, n) -categories*, arXiv:2507.00146, 2025.
- ———, *Model structures for diagrammatic (∞, n) -categories*, Algebraic & Geometric Topology (2026), to appear.
- C. Chanavat, *Homotopy theory of stricter n -categories*, arXiv:2509.26563, 2025.
- ———, *A strengthened (∞, n) -categorical pasting theorem.*, arXiv:2603.22242, 2026.
- A. Hadzihanovic, *Combinatorics of higher-categorical diagrams*, London Mathematical Society Lecture Note Series, 2024, to appear.